Ruolin Li

AI, Autonomy, and Mobility Research Leader

ruolinl@usc.edu (213) 821-4101
ruolinli.me Google Scholar
Los Angeles, California

Executive Profile

Research leader working at the intersection of **agentic AI, autonomous systems, game theory, and future mobility**. I build decision and coordination technologies for systems in which autonomous agents, companies, infrastructure, and people interact strategically. As a USC professor and lead PI of a four-year NSF FIRE initiative, I translate rigorous models into programs spanning mixed-autonomy transportation, mobility markets, wildfire evacuation, and resilient multi-agent AI. I bring experience forming interdisciplinary teams, shaping fundable research agendas, mentoring technical talent, and communicating complex systems to academic, industry, and public-sector partners.

Impact at a Glance

$2M	17	20+	12
Lead-PI NSF initiative	Research outputs	Invited talks	Students mentored

Signature Programs and Capabilities

EVAC-AI: Agentic AI for Wildfire Evacuation Lead PI | NSF FIRE #2623936 | 2026–2030

- Leads an approximately $2M, four-year USC program integrating multi-agent AI, control, transportation, and community resilience.
- Sets the technical vision and translates it into research thrusts, testbeds, stakeholder engagement, and regional-impact strategy.

Strategic Mobility and Autonomous-Vehicle Markets Principal Investigator | NCST / U.S. DOT

- Develops equilibrium and decision-support tools for networks shared by human drivers and competing autonomous-mobility operators.
- Connects market incentives, fleet operations, traffic control, and public policy through game theory and optimization.

Societally-Aware Interactive Autonomous Systems (SIAS) Lab Founder and Director

- Builds autonomous systems that account for strategic human and institutional behavior while improving collective outcomes.
- Research portfolio spans mixed autonomy, multi-agent coordination, robust learning, mechanism design, and mobility infrastructure.

Selected technical capabilities: multi-agent and agentic AI; game-theoretic control; optimization and equilibrium computation; autonomous-vehicle operations; mobility markets; simulation and digital twins; AI safety and resilience; research strategy and proposal leadership.

Selected Recognition

Best Paper Award, 8th Bridging Transportation Researchers Conference (MAPLE paper)	2026
Finalist, IEEE ITSS Best Dissertation Award	2024
WiSE Gabilan Assistant Professorship, University of Southern California	2024
Rising Star in Mechanical Engineering, Stanford University	2022
Rising Star in Civil and Environmental Engineering, MIT	2021

Selected Technical Contributions

- **MAPLE - macro–micro aligned traffic control:** developed a Pigouvian lane-control framework that translates network-level efficiency objectives into implementable vehicle-level incentives for mixed-autonomy traffic. Best Paper Award, BTR 2026.
- **Strategic autonomous vehicles at bottlenecks:** models how human drivers decide whether to remain steadfast or bypass congestion, then designs autonomous-vehicle interventions that reshape these incentives even at limited market penetration.
- **Interactive mobility markets:** studies equilibrium among travelers, competing autonomous fleets, and infrastructure operators under waiting-time, service, and road-capacity constraints.
- **Realistic AV testing:** contributed to generative methods for producing safety-critical extreme driving behaviors, improving the realism and coverage of autonomous-vehicle evaluation; published at IEEE ICRA 2025.
- **Traffic sensing and infrastructure design:** established structural properties that make network-scale sensor placement more computationally tractable and interpretable.

Selected Research Portfolio

Mixed-Autonomy Control and Infrastructure. Designs lanes, signals, incentives, and control policies that use the capabilities of connected and autonomous vehicles without assuming immediate fleet-wide adoption.

Multi-Agent AI and Controlled Emergence. Develops local incentives, communication structures, and verification methods for large collectives of AI agents that must coordinate without a central orchestrator.

Mobility Operations and Market Design. Connects traveler choice, fleet strategy, pricing, congestion, and service quality to understand how competing robotaxi or AV operators reshape transportation networks.

Resilient Decision Systems. Extends autonomy and network-control tools to wildfire evacuation and other high-consequence settings requiring uncertainty awareness, public coordination, and robust operation.

Leadership, Partnerships, and External Engagement

- Builds collaborations across civil engineering, computer science, robotics, operations research, public agencies, and industry.
- Vice Chair for Social Media, IFAC Technical Committee 9.2 on Social Impact of Automation; organizer of the IFAC Workshop on Cyber-Physical Human Systems.
- Reviewer for NSF programs in control, cyber-physical systems, smart communities, and data-enabled engineering; reviewer for leading control, robotics, AI, and transportation venues.
- Delivered 20+ invited talks at venues including Google Research, ETH, Stanford CARS, USC ISI, UCLA, UC Irvine, UC Riverside, and major professional workshops.
- Designed and teaches USC's graduate course *Game Theory for Interactive Autonomy*; mentors researchers across CEE, CS, EE, data science, and urban planning.

Partnership and Advisory Value

Research strategy. Identifies high-leverage problems, converts emerging technology trends into coherent R&D programs, and builds competitive multi-investigator proposals.

Technical diligence. Evaluates autonomy and AI concepts across scientific novelty, system feasibility, operational constraints, and credible pathways to deployment.

Coalition building. Connects researchers, industry teams, agencies, and community stakeholders around shared testbeds, data, and measurable outcomes.

Experience and Education

University of Southern California 2024–present
WiSE Gabilan Assistant Professor, Civil and Environmental Engineering; Assistant Professor, Computer Science (courtesy).

Stanford University 2023–2024
CARS Postdoctoral Fellow, Aeronautics and Astronautics; advisor: Marco Pavone.

University of California, Berkeley 2016–2023
Ph.D. and M.S., Mechanical Engineering (Controls); advisor: Roberto Horowitz.

Jilin University 2012–2016
B.S., Automotive Engineering, summa cum laude.

Selected Publications

- K. Wang, G. Huang, Z. Wang, J. Li, and **R. Li**. "MAPLE: Macro–Micro Aligned Pigouvian Lane Control for Mixed-Autonomy Traffic Efficiency." Under review, 2026. **Best Paper Award, BTR**.
- K. Wang, H. He, and **R. Li**. "When Altruism Meets Autonomy: Managing Bottleneck Congestion with Strategic Autonomous Vehicles." Under review, 2026.
- R. Dyro, M. Foutter, **R. Li**, et al. "Realistic Extreme Behavior Generation for Improved AV Testing." *IEEE ICRA*, 2025.
- **R. Li**, N. Mehr, and R. Horowitz. "Submodularity of Optimal Sensor Placement Problems for Traffic Networks." *Transportation Research Part B*, 2023.
- **R. Li** and R. Horowitz. "Dynamic Routing and Queueing for Human-Driven and Autonomous Vehicles with Responsive Signal Control." *ACC*, 2022.
- **R. Li**, P. N. Brown, and R. Horowitz. "A Highway Toll Lane Framework that Unites Autonomous Vehicles and High-Occupancy Vehicles." *IEEE ITSC*, 2021.
- N. Mehr, **R. Li**, and R. Horowitz. "A Game-Theoretic Macroscopic Model of Lane Choices at Traffic Diverges with Applications to Mixed-Autonomy Networks." *Transportation Research Part B*, 2020.

Selected Speaking and Knowledge Leadership

Delivered more than 20 invited talks and seminars at Google Research, ETH Autonomy Talks, Stanford CARS, USC Information Sciences Institute, UCLA, UC Irvine, UC Riverside, and major control, optimization, and transportation workshops. Topics span societally-aware autonomy, strategic AV control, interactive mobility systems, and the future of intelligent transportation.

Teaching and Talent Development

Designed USC's graduate course *Game Theory for Interactive Autonomy*, connecting equilibrium models, mechanism design, learning, and control with autonomous-system applications. Has mentored 12 doctoral, master's, and undergraduate researchers across civil engineering, computer science, electrical engineering, data science, and urban planning. The student-led MAPLE paper received a 2026 BTR Best Paper Award.

Full funding, publication, teaching, advising, and professional-service record available in the academic CV and at ruolinli.me.